

MS4215: Lab 1

Spotify Track Energy

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Introduction

The dataset `lab1_dataset_spotify.csv` contains information on 114,000 tracks from Spotify, covering a wide range of genres. The variables include:

Variable	Description
<code>track_genre</code>	Genre of the track
<code>energy</code>	Intensity and activity of the track (0–1)
<code>loudness</code>	Overall loudness in decibels (dB)
<code>acousticness</code>	Likelihood the track is acoustic (0–1)
<code>valence</code>	Musical positiveness of the track (0–1)
<code>danceability</code>	How suitable the track is for dancing (0–1)
<code>tempo</code>	Estimated tempo in beats per minute (BPM)
<code>speechiness</code>	Presence of spoken words in the track (0–1)
<code>popularity</code>	Track popularity score (0–100)
<code>explicit</code>	Whether the track has explicit lyrics (TRUE/FALSE)
<code>duration_ms</code>	Duration of the track in milliseconds

The outcome variable for this lab is **energy**, which measures the intensity and activity of a track. A high energy value (close to 1) indicates a fast, loud, and noisy track; a low value (close to 0) indicates a slower, quieter track.

Before attempting the questions, run the setup chunk below. This loads the required packages and draws a random sample of 3,000 tracks for analysis.

```
#install.packages("tidyverse")
#install.packages("corrplot")
library(tidyverse)
library(corrplot)

set.seed(42)

# In the line of code below, you will need to change the location to where you
# have saved the file
spotify_full <- read_csv("data/lab1_dataset_spotify.csv")
spotify      <- spotify_full |> sample_n(3000)
```

Questions

- (i) Explore the data

Use `str()` and `summary()` to inspect the dataset. What is the range of **energy** values in the sample?

- (ii) Histograms and summary statistics

Create histograms of **energy**, **loudness**, and **acousticness**. For each variable, suggest appropriate summary statistics (mean and SD, or median and IQR) and justify your choice based on the shape of the distribution.

- (iii) Scatter plot matrix and correlation

Create a scatter plot matrix and a Pearson correlation matrix for **energy**, **loudness**, **acousticness**, **valence**, **danceability**, and **tempo**.

Which predictor has the strongest linear relationship with **energy**? Describe the direction and strength of this relationship.

- (iv) Simple linear regression

Fit a simple linear regression model with **energy** as the outcome and the predictor you identified in (iii) as the explanatory variable.

- Write out the fitted model equation
- Interpret the slope and intercept in the context of Spotify tracks
- What proportion of the variation in energy does this model explain?

- (v) Multiple linear regression

Fit a multiple regression model for **energy** using **loudness**, **acousticness**, **valence**, and **danceability** as predictors.

- Interpret the partial slope for each predictor
- How does the adjusted R^2 compare to the simple regression model in (iv)?
- Are all predictors statistically significant at the 5% level?

- (vi) Residual diagnostics

Create residual plots for the multiple regression model in (v). Comment on whether the assumptions of linear regression appear to be satisfied.